

HW1

Sam Olson

Progress

- Q1: Part (d) needs work
- Q2: Draft
- Q3: Draft
- Q4: Draft
- Q5: Draft

Q1

Let X , W , and Z denote matrices.

(a)

If $\mathcal{C}(X) = \mathcal{C}(W)$, show that $P_X = P_W$, i.e., the projection matrices must be the same.

Hint: Consider $(P_W - P_X)'(P_W - P_X)$.

Proof. By Corollary 5 (4), of Lecture Notes part 1, $P_X X = X$; more generally, for any $v \in \mathcal{C}(X)$, writing $v = Xa$, $P_X v = P_X Xa = Xa = v$ (and similarly for P_W).

Since $\mathcal{C}(X) = \mathcal{C}(W)$, every column of W lies in $\mathcal{C}(X)$ and vice versa, so $P_X W = W$ and $P_W X = X$.

It then follows

$$P_W P_X = P_W X (X' X)^{-1} X' = X (X' X)^{-1} X' = P_X$$

and

$$P_X P_W = P_X W (W' W)^{-1} W' = W (W' W)^{-1} W' = P_W.$$

Using symmetry and idempotency of P_X and P_W ,

$$(P_W - P_X)'(P_W - P_X) = P_W - P_W P_X - P_X P_W + P_X = P_W - P_X - P_W + P_X = 0.$$

By Lemma 1.2, $A'A = 0 \Rightarrow A = 0$. Taking $A = P_W - P_X$, it then follows that $P_X = P_W$. //

(b)

If X is an $n \times n$ matrix of rank n , show $P_X = I_n$ (i.e., the $n \times n$ identity matrix).

Proof.

Since X is $n \times n$ of rank n , X^{-1} exists (as X is full-rank).

By Lemma 2.3, $\text{rank}(X'X) = \text{rank}(X) = n$, so $X'X$ is nonsingular with unique generalized inverse $(X'X)^- = (X'X)^{-1}$.

Then

$$P_X = X(X'X)^-X' = XX^{-1}(X')^{-1}X' = I_n. //$$

(c)

Let $\mathcal{C}^\perp(X)$ denote the orthogonal complement of $\mathcal{C}(X)$ (i.e., $y \in \mathcal{C}^\perp(X)$ if and only if $\underline{x}'y = 0$ for any column $\underline{x}$ of X). If a vector $\underline{b} \in \mathcal{C}(X)$ and $\underline{b} \in \mathcal{C}^\perp(X)$, show that $\underline{b}$ must equal the zero vector (this also implies that a non-zero vector in $\mathcal{C}(X)$ must be linearly independent from a non-zero vector in $\mathcal{C}^\perp(X)$).

Hint: Write $\underline{b} = X\underline{a}$ and consider $\underline{b}'\underline{b}$.

Proof.

Since $\underline{b} \in \mathcal{C}(X)$, it follows that $\underline{b} = X\underline{a}$ for some $\underline{a}$.

Since $\underline{b} \in \mathcal{C}^\perp(X)$, it follows that $X'\underline{b} = \underline{0}$.

Then $\underline{b}'\underline{b} = (X\underline{a})'\underline{b} = \underline{a}'X'\underline{b} = 0$. However, since $\underline{b}'\underline{b} = \sum_i b_i^2 = 0$, it follows that $b_i = 0$, for all i , and we conclude that $\underline{b} = \underline{0}$. //

(d)

Let X be an $n \times p$ matrix of rank $r \geq 1$, and Z be an $n \times (n - r)$ matrix with columns formed by $n - r$ linearly independent vectors in $\mathcal{C}^\perp(X)$. Show that $P_Z = I_n - P_X$.

Hint: Put r linearly independent columns of X along with the columns of Z into one $n \times n$ matrix $\tilde{X}$ (if $n = r$, technically Z is just a zero vector). Compute $P_{\tilde{X}}$ and use (a)-(b).

Proof.

Let X_1 ($n \times r$) consist of r linearly independent columns of X , so $\mathcal{C}(X_1) = \mathcal{C}(X)$ and $P_{X_1} = P_X$ by (a) given above. Then, let $\tilde{X} = (X_1 \ Z)$.

Let us start by considering the degenerate case, $r = n$. If Z has zero columns, $\mathcal{C}^\perp(X) = \{\underline{0}\}$, $P_X = I_n$, and $P_Z = I_n - P_X = 0$ holds trivially.

Let us then consider the other case.

Nonsingularity: X_1 's columns span $\mathcal{C}(X)$ ($\dim r$), Z 's span $\mathcal{C}^\perp(X)$ ($\dim n - r$). By (c), $\mathcal{C}(X) \cap \mathcal{C}^\perp(X) = \{\underline{0}\}$, so all n columns of $\tilde{X}$ are independent: $\text{rank}(\tilde{X}) = n$.

Block structure: Columns of Z lie in $\mathcal{C}^\perp(X)$, so $X_1'Z = 0 = Z'X_1$, giving

$$\tilde{X}'\tilde{X} = \begin{pmatrix} X_1'X_1 & 0 \\ 0 & Z'Z \end{pmatrix}, \quad (\tilde{X}'\tilde{X})^{-1} = \begin{pmatrix} (X_1'X_1)^{-1} & 0 \\ 0 & (Z'Z)^{-1} \end{pmatrix},$$

where both diagonal blocks are nonsingular by Lemma 2.3. Hence

$$P_{\tilde{X}} = \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}' = X_1(X_1'X_1)^{-1}X_1' + Z(Z'Z)^{-1}Z' = P_{X_1} + P_Z = P_X + P_Z.$$

Since $\tilde{X}$ is nonsingular, (b) gives $P_{\tilde{X}} = I_n$, so $P_Z = I_n - P_X$. //

(e)

For a vector $\underline{y}$, show that $P_X \underline{y} = 0$ if and only if $\underline{y} \in \mathcal{C}^\perp(X)$.

Hint: use (d) and multiply $I = I - P_X + P_X$ by $\underline{y}$

Proof.

($\Leftarrow$) Suppose $\underline{y} \in \mathcal{C}^\perp(X) = \mathcal{C}(Z)$.

Since $\underline{y} \in \mathcal{C}(Z)$, it follows that $P_Z \underline{y} = \underline{y}$.

By (d), $I = P_X + P_Z$. Multiplying by $\underline{y}$ gives,

$$\underline{y} = I\underline{y} = P_X \underline{y} + P_Z \underline{y} = P_X \underline{y} + \underline{y} \implies P_X \underline{y} = \underline{0}.$$

($\Rightarrow$) If $P_X \underline{y} = \underline{0}$, then by (d), $\underline{y} = I\underline{y} = P_X \underline{y} + P_Z \underline{y} = P_Z \underline{y} \in \mathcal{C}(Z) = \mathcal{C}^\perp(X)$. //

(f)

Show that $P_X + P_W$ is a projection matrix if and only if $X'W = 0$ (the columns of X and W must be orthogonal).

Hint: If $P_X + P_W$ is a projection matrix, one can show $P_X P_W = -P_W P_X$ must hold; this relation implies that any vector $\underline{y}$ that is in both $\mathcal{C}(X)$ and $\mathcal{C}(W)$ must be zero (why?). Further, for $M = P_X P_W$, the relation also gives $M'M = P_X P_W P_W$ and $M'M = P_W P_X P_W$, meaning any column of $M'M$ belongs to $\mathcal{C}^\perp(X)$ and to $\mathcal{C}^\perp(W)$.

Preliminary fact (used in (f), (g)): M is symmetric and idempotent $\iff M = P_A$ for some A . ($\Rightarrow$: Corollary 5. $\Leftarrow$: if $M' = M$, $M^2 = M$, then $M \cdot M \cdot M = M$, so M is a generalized inverse of $M'M = M$, giving $P_M = M(M'M)^- M' = M^3 = M$.) So “ $P_X + P_W$ is a projection matrix” $\iff$ “ $P_X + P_W$ is idempotent” (symmetry is automatic).

Proof.

($\Leftarrow$) If $X'W = 0$ (so also $W'X = 0$), then $P_X P_W = X(X'X)^- X'W(W'W)^- W' = 0$ and $P_W P_X = 0$, so $(P_X + P_W)^2 = P_X + P_W$: idempotent, hence a projection matrix.

($\Rightarrow$) If $P_X + P_W$ is a projection matrix, idempotency gives

$$P_X + P_X P_W + P_W P_X + P_W = P_X + P_W \implies P_X P_W = -P_W P_X. \quad (*)$$

Step 1: $\mathcal{C}(X) \cap \mathcal{C}(W) = \{0\}$. If $\underline{y} \in \mathcal{C}(X) \cap \mathcal{C}(W)$, then $P_X \underline{y} = \underline{y} = P_W \underline{y}$ (reproduction property). Applying (*) to $\underline{y}$: $P_X P_W \underline{y} = P_X \underline{y} = \underline{y}$, while $-P_W P_X \underline{y} = -P_W \underline{y} = -\underline{y}$. Since these are equal by (*), $\underline{y} = -\underline{y}$, so $\underline{y} = \underline{0}$.

Step 2: columns of $M'M$, where $M = P_X P_W$, lie in both $\mathcal{C}(X)$ and $\mathcal{C}(W)$. Using $P_X^2 = P_X$,

$$M'M = P_W P_X P_X P_W = P_W P_X P_W = P_W (P_X P_W) = P_W M,$$

so $\mathcal{C}(M'M) \subseteq \mathcal{C}(P_W) = \mathcal{C}(W)$. Separately, grouping as $(P_W P_X) P_W$ and applying (*) ($P_W P_X = -P_X P_W$),

$$M'M = (P_W P_X) P_W \stackrel{(*)}{=} -P_X P_W P_W = -P_X P_W,$$

so $\mathcal{C}(M'M) \subseteq \mathcal{C}(P_X) = \mathcal{C}(X)$.

Step 3: conclude $M = 0$. By Steps 1–2, every column of $M'M$ lies in $\mathcal{C}(X) \cap \mathcal{C}(W) = \{0\}$, so $M'M = 0$. By Lemma 1.2, $M = 0$, i.e. $P_X P_W = 0$; then (*) gives $P_W P_X = 0$ too.

Right-multiplying $P_X P_W = 0$ by W and using $P_W W = W$ gives $P_X W = 0$; left-multiplying by X' and using $X' P_X = X'$ gives $X' W = X' P_X W = 0$. //

(g)

Show that $P_X - P_W$ is a projection matrix if and only if $\mathcal{C}(W) \subset \mathcal{C}(X)$.

Hint: If $P_X - P_W$ is a projection matrix then $(I - P_X)P_W = -P_W(I - P_X)$ must hold; then the proof is kind of the same as in (f), using that (from (d)) $P_Z = I_n - P_X$ holds, where Z is a matrix with linearly independent columns that span $\mathcal{C}^\perp(X)$.

Proof.

($\Leftarrow$) If $\mathcal{C}(W) \subseteq \mathcal{C}(X)$, then since Z 's columns span $\mathcal{C}^\perp(X)$, they're orthogonal to $\mathcal{C}(X) \supseteq \mathcal{C}(W)$, so $Z'W = 0$. By (f), $P_Z P_W = P_W P_Z = 0$ and $P_Z + P_W$ is symmetric idempotent. Then $P_X - P_W = (I - P_Z) - P_W = I - (P_Z + P_W)$, and for symmetric idempotent Q , $I - Q$ is symmetric idempotent, so $P_X - P_W$ is a projection matrix.

($\Rightarrow$) If $P_X - P_W$ is a projection matrix, idempotency gives

$$P_X - P_X P_W - P_W P_X + P_W = P_X - P_W \implies P_X P_W + P_W P_X = 2P_W.$$

Substituting $P_X = I - P_Z$:

$$(I - P_Z)P_W + P_W(I - P_Z) = 2P_W \implies P_Z P_W = -P_W P_Z,$$

which is exactly relation (*) from (f) with Z in place of X . The identical computation as in (f) then gives $Z'W = 0$, i.e. $\mathcal{C}(W) \subseteq \mathcal{C}^\perp(Z)$.

Since $\mathcal{C}(X) \subseteq \mathcal{C}^\perp(Z)$ by construction and $\dim \mathcal{C}^\perp(Z) = n - (n - r) = r = \dim \mathcal{C}(X)$, equal dimension plus containment gives $\mathcal{C}^\perp(Z) = \mathcal{C}(X)$. Hence $\mathcal{C}(W) \subseteq \mathcal{C}(X)$. //

2.

Suppose that $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and that $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$.

(a)

Show that $AA^-C = C$ and $DA^-A = D$.

Proof.

Since $\mathcal{C}(C) \subseteq \mathcal{C}(A)$, Lemma 1.6(i) gives $C = AF$ for some matrix F . Then

$$AA^-C = AA^-(AF) = (AA^-A)F = AF = C,$$

using $AA^-A = A$ (the defining property of a generalized inverse).

Since $\mathcal{R}(D) \subseteq \mathcal{R}(A)$, Lemma 1.6(ii) gives $D = GA$ for some matrix G . Then

$$DA^-A = (GA)A^-A = G(AA^-A) = GA = D. //$$

(b)

Using (a), show that

$$\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} \begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix},$$

where $Q = B - DA^-C$, and I denotes the identity matrix of appropriate dimensions.

Proof.

First compute $\begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix}$ blockwise:

- top-left: $AI + C \cdot 0 = A$
- top-right: $A(-A^-C) + CI = -AA^-C + C = -C + C = 0$ (by (a))
- bottom-left: $DI + B \cdot 0 = D$
- bottom-right: $D(-A^-C) + BI = B - DA^-C = Q$

$$\begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ D & Q \end{bmatrix}.$$

Now left-multiply by $\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$:

- top-left: $IA + 0 \cdot D = A$
- top-right: $I \cdot 0 + 0 \cdot Q = 0$
- bottom-left: $-DA^-A + ID = -D + D = 0$ (by (a))
- bottom-right: $-DA^- \cdot 0 + IQ = Q$

$$\begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} \begin{bmatrix} A & 0 \\ D & Q \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix}. //$$

(c)

Using (b) with the fact that the two matrices defined by “ I on the diagonals” are square and invertible in (b), show that

$$\begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$$

is a generalized inverse of

$$M \equiv \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

Proof.

Let $L = \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix}$, $R = \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix}$, and $\tilde{M} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix}$. Direct multiplication confirms

$$L^{-1} = \begin{bmatrix} I & 0 \\ DA^- & I \end{bmatrix}, \quad R^{-1} = \begin{bmatrix} I & A^-C \\ 0 & I \end{bmatrix}$$

(e.g. $LL^{-1} = \begin{bmatrix} I & 0 \\ -DA^- + DA^- & I \end{bmatrix} = I$, and similarly for R), so L, R are invertible.

By (b), $LMR = \tilde{M}$, so $M = L^{-1}\tilde{M}R^{-1}$.

Let $G = R \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} L$. Then

$$MGM = (L^{-1}\tilde{M}R^{-1}) \left(R \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} L \right) (L^{-1}\tilde{M}R^{-1}) = L^{-1}\tilde{M} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \tilde{M}R^{-1},$$

since $R^{-1}R = I$ and $LL^{-1} = I$. Now

$$\tilde{M} \begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \tilde{M} = \begin{bmatrix} AA^-A & 0 \\ 0 & QQ^-Q \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix} = \tilde{M},$$

using $AA^-A = A$ and $QQ^-Q = Q$. Hence $MGM = L^{-1}\tilde{M}R^{-1} = M$, so G is a generalized inverse of M . //

(d)

Show that the above generalized inverse matches that given in (i) of Theorem 1.4.

Proof.

First,

$$\begin{bmatrix} A^- & 0 \\ 0 & Q^- \end{bmatrix} \begin{bmatrix} I & 0 \\ -DA^- & I \end{bmatrix} = \begin{bmatrix} A^- & 0 \\ -Q^-DA^- & Q^- \end{bmatrix}.$$

Then

$$R \begin{bmatrix} A^- & 0 \\ -Q^-DA^- & Q^- \end{bmatrix} = \begin{bmatrix} I & -A^-C \\ 0 & I \end{bmatrix} \begin{bmatrix} A^- & 0 \\ -Q^-DA^- & Q^- \end{bmatrix} = \begin{bmatrix} A^- + A^-CQ^-DA^- & -A^-CQ^- \\ -Q^-DA^- & Q^- \end{bmatrix},$$

computing entrywise: top-left = $A^- + (-A^-C)(-Q^-DA^-) = A^- + A^-CQ^-DA^-$; top-right = $(-A^-C)Q^- = -A^-CQ^-$; bottom-left = $-Q^-DA^-$; bottom-right = Q^- .

This is exactly the matrix given in Theorem 1.4(i). //

(e)

Show that the conditions imposed on the matrices B , C , and D in part (ii) of Theorem 1.4 are satisfied if there exists a matrix $Z = (Z_1, Z_2)$ for which

$$M = Z'Z = (Z_1, Z_2)'(Z_1, Z_2) = \begin{bmatrix} Z_1'Z_1 & Z_1'Z_2 \\ Z_2'Z_1 & Z_2'Z_2 \end{bmatrix} = \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

Proof.

By Lemma 1.4 (applied with $X = Z_2$), $Z_2'Z_2(Z_2'Z_2)^-Z_2' = Z_2'$, so $Z_2' = (Z_2'Z_2)[(Z_2'Z_2)^-Z_2']$, which by Lemma 1.6(i) gives $\mathcal{C}(Z_2') \subseteq \mathcal{C}(Z_2'Z_2)$. Conversely, $Z_2'Z_2 = Z_2' \cdot Z_2$ trivially gives $\mathcal{C}(Z_2'Z_2) \subseteq \mathcal{C}(Z_2')$ (Lemma 1.6(i) again). Hence

$$\mathcal{C}(Z_2') = \mathcal{C}(Z_2'Z_2) = \mathcal{C}(B).$$

Since $D = Z_2'Z_1$, each column of D is Z_2' times a column of Z_1 , so by Lemma 1.6(i), $\mathcal{C}(D) \subseteq \mathcal{C}(Z_2') = \mathcal{C}(B)$. This is exactly the first condition of Theorem 1.4(ii).

For the second condition, note $C = Z_1'Z_2$ so $C' = (Z_1'Z_2)' = Z_2'Z_1 = D$, and $B' = (Z_2'Z_2)' = Z_2'Z_2 = B$. Then

$$\mathcal{R}(C) = \mathcal{C}(C') = \mathcal{C}(D) \subseteq \mathcal{C}(B) = \mathcal{C}(B') = \mathcal{R}(B),$$

which is exactly the second condition of Theorem 1.4(ii). //

3.

For each of the following statements, indicate whether the statement is true or false, and explain.

(a)

If $Ax = y$ has a unique solution, then A must be a nonsingular matrix.

False.

Let A be an $n \times m$ matrix and suppose $A\mathbf{x} = \mathbf{y}$ is consistent, with particular solution $\mathbf{x}_0$. The full solution set is $\mathbf{x}_0 + \mathcal{N}(A)$, so the solution is unique if and only if $\mathcal{N}(A) = \{\mathbf{0}\}$, i.e., A has full column rank. This is purely a rank condition and has nothing intrinsically to do with A being square — indeed, “nonsingular” is a term defined only for square matrices. As a counterexample, let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

a 3×2 matrix of full column rank. For any $\mathbf{y}$ in the column space of A — say $\mathbf{y} = A(2, -1)^\top$ — the system $A\mathbf{x} = \mathbf{y}$ has the unique solution $\mathbf{x} = (2, -1)^\top$, even though A is not square and so cannot properly be called nonsingular at all.

(b)

If G is a generalized inverse of A , then $GA = I$, where I denotes the identity matrix of appropriate dimensions.

False.

The defining property of a generalized inverse is only $AGA = A$; nothing in this equation forces $GA = I$. As a counterexample, let $A = \text{diag}(1, 0)$, a symmetric, singular, idempotent matrix ($A^2 = A$), and take $G = I$. Then $AGA = A \cdot I \cdot A = A^2 = A$, so $G = I$ is a valid generalized inverse of A . However, $GA = IA = A \neq I$, since A is singular. This single example is enough to show that a valid generalized inverse need not satisfy $GA = I$.

(c)

If A has a generalized inverse G such that $GA = I$, then A must be a (square) nonsingular matrix.

False.

The condition $GA = I_m$ forces A (of size $n \times m$) to have full column rank, but this is again a rank condition, not a requirement that $n = m$. Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

a 3×2 matrix of full column rank, so that $A^\top A$ is 2×2 and nonsingular. Setting $G = (A^\top A)^{-1} A^\top$ gives $GA = (A^\top A)^{-1} A^\top A = I_2$ and $AGA = A(A^\top A)^{-1} A^\top A = A$, so G is a genuine generalized inverse of A satisfying $GA = I$. Yet A itself is 3×2 — rectangular, not square — and therefore cannot be nonsingular. (This G is, in fact, the Moore–Penrose pseudoinverse of A in the full-column-rank case, the same object used to prove part (e).)

(d)

If A is a symmetric matrix, then all its generalized inverses are also symmetric.

False.

Symmetry of A constrains A , not G : the equation $AGA = A$ generally admits many solutions G , most of which are not symmetric. Let $A = \text{diag}(1, 0)$, a symmetric matrix, and take

$$G = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Direct computation gives $AGA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = A$, so G is a valid generalized inverse of A , but $G \neq G^\top$. This counterexample shows that not every generalized inverse of a symmetric matrix is itself symmetric.

(e)

If A is a symmetric matrix, then it has at least one symmetric generalized inverse.

True.

Unlike the previous parts, this is an existence claim, so a counterexample cannot settle it — a constructive proof is required instead. Because A is symmetric, the spectral theorem guarantees an orthogonal matrix P (with $P^\top P = PP^\top = I$) and a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ of eigenvalues such that $A = PDP^\top$. Define D^- to be the diagonal matrix whose i th entry is $1/\lambda_i$ when $\lambda_i \neq 0$ and 0 when $\lambda_i = 0$, and set $G = PD^-P^\top$. Since D^- is diagonal, $G^\top = P(D^-)^\top P^\top = PD^-P^\top = G$, so G is symmetric. Moreover, since $P^\top P = I$ and $DD^-D = D$ entrywise regardless of whether λ_i is zero,

$$AGA = (PDP^\top)(PD^-P^\top)(PDP^\top) = P(DD^-D)P^\top = PDP^\top = A.$$

Thus $G = PD^-P^\top$ is symmetric and satisfies $AGA = A$, proving that every symmetric matrix has at least one symmetric generalized inverse.

4.

For each of the following questions, give, if possible, a 3×3 **nondiagonal** matrix A that has the stated property.

(a)

A is a generalized inverse of A .

The condition A is a generalized inverse of itself means $A \cdot A \cdot A = A$, i.e. $A^3 = A$. This holds for any idempotent matrix, since $A^2 = A$ implies $A^3 = A^2 \cdot A = A \cdot A = A^2 = A$. Take

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

a nondiagonal matrix with $A^2 = A$. Consequently $A^3 = A$, so A is a generalized inverse of itself.

(b)

A' is a generalized inverse of A .

We need $AA^\top A = A$, which holds whenever $A^\top = A^{-1}$, i.e. whenever A is orthogonal. A nondiagonal orthogonal matrix is easy to produce: let A be the 3×3 cyclic permutation matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Since $A^\top A = I$, we have $A^\top = A^{-1}$, and so $AA^\top A = AA^{-1}A = A$. Note $A^\top \neq A$ here (the transpose is the inverse permutation), which nicely illustrates that A' need not equal A to serve as its generalized inverse.

(c)

The identity matrix I is a generalized inverse of A .

The condition is $AIA = A^2 = A$, i.e. A is idempotent — exactly the same requirement used in part (a). The matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

from part (a) again works, since $A^2 = A$ directly gives $AIA = A$. It's worth noting that idempotency ($A^2 = A$) is strictly stronger than the condition $A^3 = A$ needed in (a): every idempotent matrix satisfies (a), but not every matrix satisfying (a) is idempotent (a nondiagonal reflection, with $A^2 = I \neq A$ but $A^3 = A$, would satisfy (a) without satisfying (c)).

(d)

Every 3×3 matrix is a generalized inverse of A .

Not possible. Requiring $AGA = A$ for *every* 3×3 matrix G is an extremely strong condition. Taking $G = 0$ in particular forces $A \cdot 0 \cdot A = 0 = A$, so A must be the zero matrix; and conversely, if $A = 0$ then $AGA = 0 = A$ trivially holds for any G . Hence $A = 0$ is the *unique* matrix with this property. But the zero matrix has all off-diagonal entries equal to zero, so it is itself a diagonal matrix — there is no nondiagonal 3×3 matrix for which every matrix serves as a generalized inverse.

(e)

Every generalized inverse of A is nonsingular.

This property holds if and only if A is nonsingular. If A is nonsingular, the equation $AGA = A$ has the unique solution $G = A^{-1}$, which is of course nonsingular.

Conversely, suppose A is singular, with $\text{rank}(A) = r < 3$. By Theorem 1.1, there exist nonsingular 3×3 matrices P, Q and a nonsingular $r \times r$ diagonal matrix D with $PAQ = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$, and $G = Q \begin{pmatrix} D^{-1} & F \\ H & B \end{pmatrix} P$ is a generalized inverse of A for *any* choice of F, H, B . Taking $F = H = B = 0$, the matrix $G_0 = Q \begin{pmatrix} D^{-1} & 0 \\ 0 & 0 \end{pmatrix} P$ satisfies

$$\text{rank}(G_0) = \text{rank} \begin{pmatrix} D^{-1} & 0 \\ 0 & 0 \end{pmatrix} = r < 3,$$

since pre- and post-multiplying by the nonsingular matrices P, Q preserves rank. So G_0 is a singular generalized inverse of A , ruling the property out for any singular A . A nondiagonal nonsingular example where the property *does* hold:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \det(A) = 1 \neq 0.$$

Its unique generalized inverse, A^{-1} , is nonsingular by construction.

5.

For any matrix A , let B and C be any two matrices that satisfy $A'AB = A'$ and $AA'C' = A$ (the existence of such B and C follows from Corollary 2 to Lemma 1.4). Let $G = CAB$. Show that G satisfies the following four conditions:

(i)

$$AGA = A$$

Proof.

We invoke Corollary 1 to Lemma 1.5: for any matrix X and any matrix B satisfying $X'XB = X'$, it holds that $XBX = X$.

(1) Take $X = A$: given $A'AB = A'$, with $X = A$, it then follows that $X'XB = X'$.

The corollary gives $XBX = X$, i.e.

$$ABA = A. \tag{1}$$

(2) Take $X = A'$: given $AA'C' = A$, substitute B with C' :

$$X'XC' = AA'C' = AA'C' = A = X',$$

So $X'XC' = X'$ holds, and application of the corollary gives $XC'X = X$, i.e., $A'C'A' = A'$.

Now transpose both sides (using $(PQR)' = R'Q'P'$ with $P = A', Q = C', R = A'$):

$$(A'C'A')' = (A')'(C')'(A')' = ACA,$$

Using $(A')' = A$, we then have

$$ACA = A. \tag{2}$$

(The rest of the proof)

Now, with $G = CAB$,

$$AGA = A(CAB)A = (ACA)(BA) = A(BA) = ABA = A. //$$

(ii)

$$GAG = G$$

Proof.

Using (1) from part (i),

$$GA = (CAB)A = C(ABA) = CA, \tag{3}$$

and similarly using (2),

$$AG = A(CAB) = (ACA)B = AB. \tag{4}$$

Taking everything together,

$$GAG = (GA)G = (CA)(CAB) = C(ACA)B = CAB = G. //$$

(iii)

$$(GA)' = GA$$

Proof.

Take $X = A'$, so $X' = A$.

As in part (i), C' satisfies $X'XC' = AA'C' = A = X'$, so application of Corollary 4 to Lemma 1.5 gives that $XC' = A'C'$ is symmetric, i.e.:

$$(A'C')' = A'C'.$$

Since $(A'C')' = (C')'(A')' = CA$, it follows that $CA = A'C'$.

Similarly, the standard transpose rule gives $(CA)' = A'C'$.

Combining these two statements,

$$(CA)' = A'C' = CA.$$

By (3) in part (ii), $GA = CA$, so $(GA)' = (CA)' = CA = GA. //$

(iv)

$$(AG)' = AG$$

Proof.

Take $X = A$.

Since B satisfies $X'XB = A'AB = A' = X'$, Corollary 4 to Lemma 1.5 gives that $XB = AB$ is symmetric, i.e.:

$$(AB)' = AB.$$

By (4) in part (ii), $AG = AB$, so $(AG)' = (AB)' = AB = AG. //$